1. Project Goal :

The goal of our project is to explore the rise of women's international football through an interactive data story, combining historical, economic, and in-game perspectives.

We aim to highlight how the sport evolved from being restricted and under-recognized to becoming a global phenomenon, while also uncovering persistent inequalities such as prize money gaps and structural differences compared to men's football.

Our visualizations are designed to be intuitive, interactive, and narrative-driven, allowing users with little to moderate domain knowledge to explore the data and derive insights themselves. The project emphasizes storytelling through data, guiding users across time, geography, and performance metrics.

2. Outline :

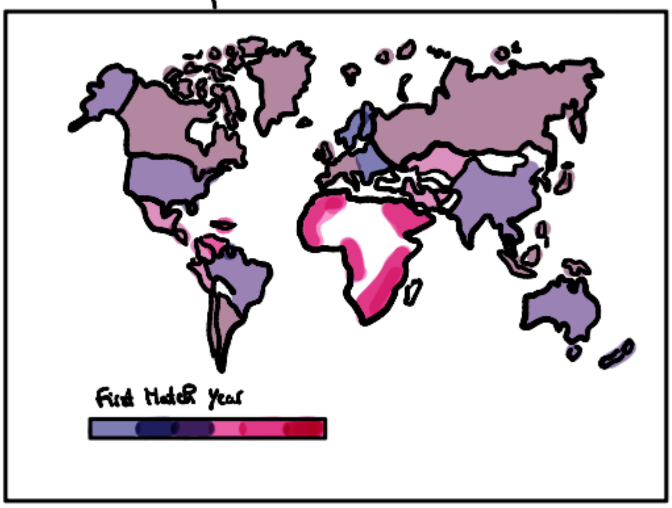
Our website is structured as a scrollytelling data story, progressing through three main themes: **historical growth → economic development → match-level insights.**

Historical Growth : *How did women's football spread globally?*

We begin with a scroll-driven world map visualization that shows when each country played its first international match.

- Countries are colored by first participation year
- The map updates dynamically as the user scrolls through time
- Key historical milestones (e.g., bans, World Cups, global expansion) are highlighted

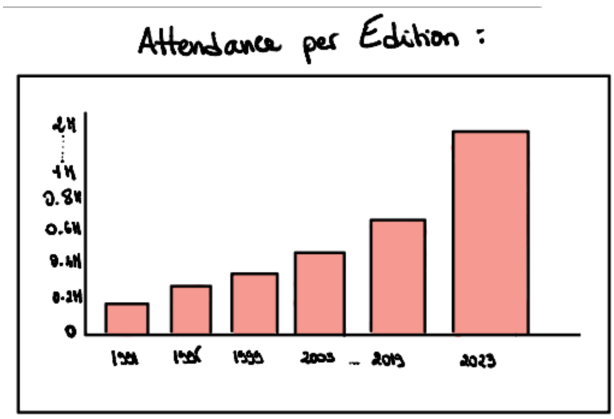
Heat Map: Historical Growth

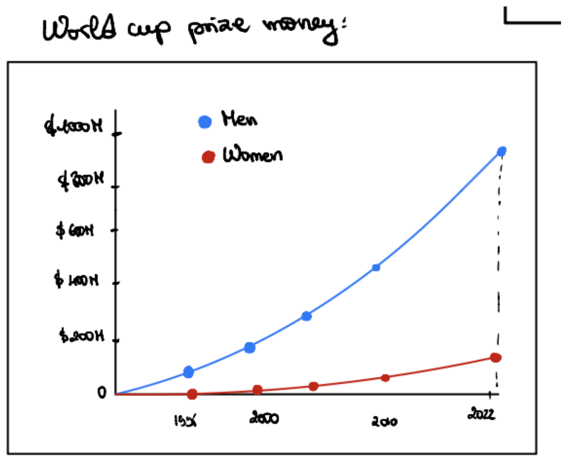


Economics : *How has the sport grown financially?*

The second section explores the economic evolution of women's football, focusing on:

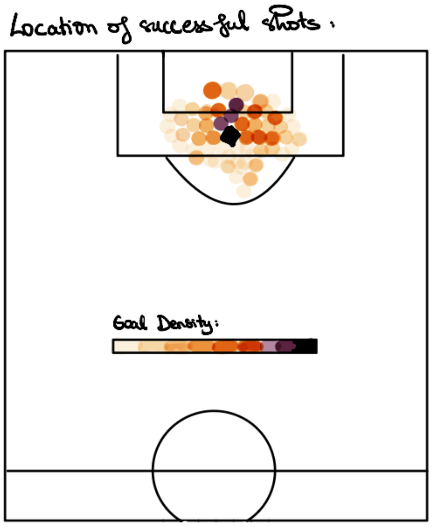
- Prize money over time (men vs women)
- World Cup attendance growth
- Number of teams participating
- Prize money gap comparison





These coordinated D3 visualizations (line charts, bar charts, and animated comparisons) highlight both growth and inequality.

Match Insights : *What happens inside the game?*



The final section focuses on game-level data, providing insights into how football is actually played:

- Goal timing heatmap (distribution across match minutes)
- Top scorers visualization
- Penalty shootout statistics
- Competitiveness trends (goal margins over time)

Together, these visualizations shift the narrative from macro (global growth) to micro (on-field dynamics).

3. Tools and Lectures :

Our data comes from the “Women’s International Football Results” dataset (11,177 matches, 1956–2025), complemented with manually collected FIFA statistics such as attendance and prize money.

Initial exploration shows strong global growth (247 teams today), increasing competitiveness (fewer extreme scorelines), and clear economic disparity (women’s $110M vs men’s $1B prize pool).

These insights directly motivate our three visualization axes: global expansion, financial growth, and match-level dynamics.

We use a consistent technical stack across all visualizations to ensure cohesion and flexibility.

Tools :

- D3.js → main visualization library (e.g., d3.geoNaturalEarth1 for map projection, d3.scaleSequential for color encoding)
- TopoJSON → world map rendering
- HTML/CSS/JavaScript → website structure and styling

Using D3 throughout allows full control over animation, transitions, and interactivity, without relying on external frameworks.

Lectures :

- Data → Used to combine the datasets from FIFA and Kaggle
- D3.js → core visualization implementation
- Interactions → hover effects, scroll-based updates
- Marks & Channels → encoding values visually
- Perception & Color → color scales and readability
- *(upcoming)* Maps lecture → improving projection and geo encoding
- *(upcoming)* Storytelling lecture → refining narrative flow

4. Breakdown :

The MVP ensures that users can understand the full narrative through three core components:

1. A static or minimally interactive world map showing first match year
2. An economics section with basic charts (prize money, attendance, teams)
3. A goal timing heatmap or simple match visualization

Prize Money Comparison:

Additional Ideas :

- Fully scroll-synchronized animated map
- Enhanced transitions and storytelling flow
- More detailed interaction (tooltips, filtering, highlighting)
- Advanced match analytics (e.g., event-based visualizations like passes/shots)
- Improved narrative polish and visual design